

Hack4Health — End-to-End Patient Process Flowcharts

End-to-End Patient Process — Flowcharts

Visualizes the current (as-is) clinic registration process from [1. Technical Problem Statement.md](#) and the proposed (to-be) flow using the ClinicPrep Assistant agent from [PLAN.md](#).

1. As-Is Process (Current, Manual)

Red nodes = duplicate/paper-based pain points. Yellow nodes = administrative work that happens sequentially at the registration counter.

Step 1 – Patient Arrives & Waits
Variable, compounds

Step 2 – Identity Verification
~1 min

Step 3 – Patient Registration
~2-3 min
(name, NRIC, DOB, address, contact typed by hand)

Step 4 – Document Interpretation
~3-5 min

If TPA patient: staff also writes attendance on a separate paper TPA form

Step 5 – Eligibility Check
~3-5 min
(CHAS / corporate insurance)

Step 6 – Screening Package Verification
~2-4 min (checkup visits only)

Patient fills 2 on-site forms:
General / Occupational Health consent form
(re-asks name, NRIC, DOB, contact, address)

Step 7 – Billing Determination
~2 min

Step 8 – Queue Number Issued
~1 min

Step 9 – Wait in Queue
Variable, compounds

Step 10 – Consultation or Screening
10-20 min

Step 11 – Medication Dispensed
~2-3 min (pharmacist re-asks allergies)

Step 12 – Payment
~5-8 min

If under insurance: staff logs into TPA portal to manually re-enter dispensed/checkup info

Admin time per patient: ~23–32 minutes, excluding the clinical visit itself (see Technical Problem Statement).

2. To-Be Process (With ClinicPrep Assistant)

Green = automated before the patient reaches the counter. Blue = must stay manual/in-person (identity + e-card verification are never automated).

Before Arrival –

Patient starts chat
(web link / kiosk / Teams)

Has a referral letter,
voucher, or e-card?

Yes

Upload / paste document

Document Parsing Action:
extract company/TPA
code, policy no.,
requested package

No / self-pay

Provide basic details

Patient Lookup Action:
pre-fill registration from
existing record

Eligibility & Package
Matching Action

Consent Questionnaire Pre-
fill
(General / Occupational
Health)

Billing Estimate Action

Handoff Summary + Queue
Ticket generated
(status: Pending
Verification)

At Clinic – In-Person (Staff)

Patient arrives, shows
queue ticket

Identity Verification
(NRIC/FIN/Passport) – IN
PERSON, MANUAL

e-Card / Insurance Card
Validation – IN PERSON,
MANUAL

Pre-filled summary
matches in-person check?

Yes

Staff confirms registration,
package & billing

No / mismatch

Escalate for manual
correction

Queue for Consultation /
Screening

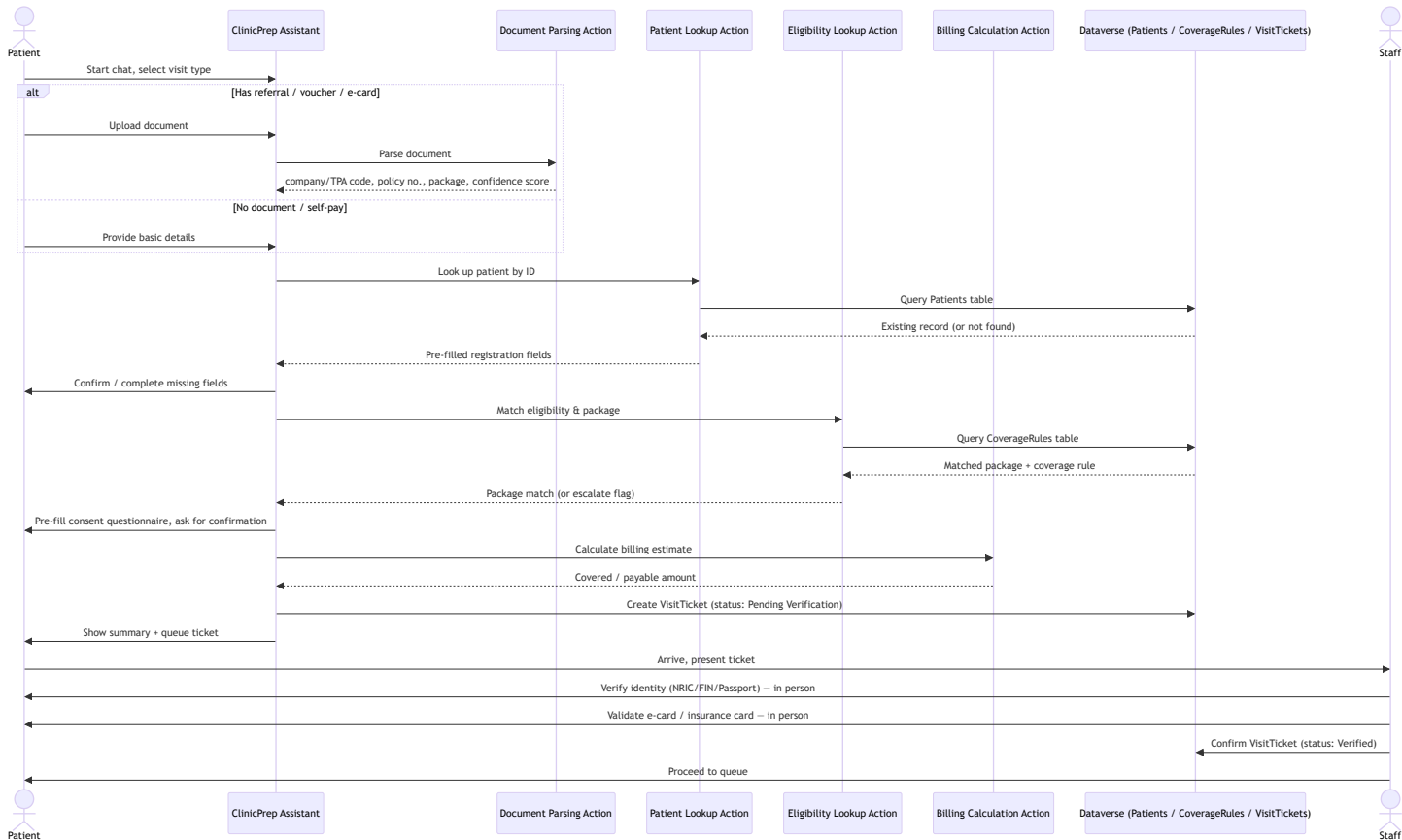
Consultation or Screening

Medication Dispensed

Payment
(pre-computed billing,
minimal re-work)

3. Sequence Diagram — Pre-Registration Interaction

Shows how the agent, its actions, Dataverse, and staff interact (maps to the topics/actions defined in [PLAN.md](#)).



4. Time Impact Summary

Stage	As-Is	To-Be
Registration + document interpretation + eligibility + package verification + billing	~10–19 min at the counter	Done before arrival via chat; staff only confirms
TPA dual paper record	Extra manual write-up (step 4)	Eliminated — captured once digitally
Consent questionnaires (General/Occupational Health)	Filled on-site, re-asking known details	Pre-filled before arrival, patient just confirms
Post-visit TPA portal re-entry	~5–8 min manual re-entry (step 12)	Same data reused from VisitTicket, no re-typing
Identity + e-card verification	~1 min, manual	Unchanged — stays manual, in person (by design constraint)

See [README.md](#) for the full Technical Track requirements and [PLAN.md](#) for the agent configuration that implements this flow.